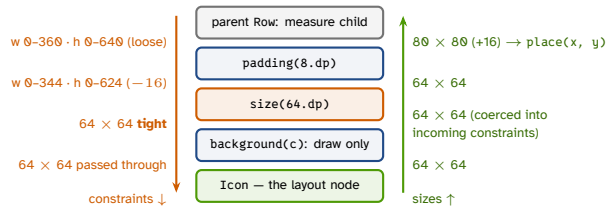
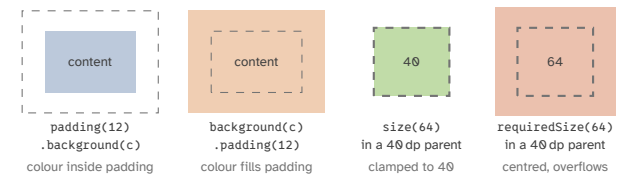


In one line: Layout is **one pass**: a parent hands each child **Constraints** (min/max width, height), the child is **measured once** and reports a **size**, the parent **places** it. A Modifier chain wraps outer → inner; each link may change the constraints going down and the size coming up — **order is behaviour**.

Icon(Modifier.padding(8.dp).size(64.dp).background(c)) in a 360 dp Row



Order = wrapping order (outer → inner)



Input too: clickable{}.padding(12) = padding tappable; reversed = not · clip before background

Measure → size → place

- MeasurePolicy gets (measurables, constraints); measure() twice on one child **throws** — single pass is enforced, so deep nesting is cheap (unlike nested RelativeLayout).
- Tight** = min = max (Constraints.fixed); **loose** = min 0; **unbounded** = max Infinity (in a scrollable).
- layout(w, h) { p.placeRelative(x, y) } — placeRelative mirrors in RTL, place does not. A size outside the constraints is coerced + centred.

Size modifiers & intrinsics

- size = a *preference* coerced into incoming constraints; the first tight one wins: size(100).size(50) = **100**. requiredSize ignores them.
- fillMaxWidth(f): min = max = f · max. wrapContentSize(): min → 0 + align — a child smaller than a tight parent.
- weight(w): unweighted children measured first, the rest split by weight (fill = false = a cap). Box: matchParentSize() takes the size *after* siblings set it; fillMaxSize() helps set it.
- Row(Modifier.height(IntrinsicSize.Min)) asks children minIntrinsicHeight *before* measuring, so a fillMaxHeight() divider matches the tallest text. A query, not a measure — but a second walk. A custom Layout's default intrinsics are **approximated**: override them.

Example — a custom Layout

```
@Composable fun Stack(modifier: Modifier = Modifier,
    content: @Composable () -> Unit) =
    Layout(content, modifier) { measurables, constraints ->
        val loose = constraints.copy(minWidth = 0, minHeight = 0)
        val ps = measurables.map { it.measure(loose) } // once each
        val w = (ps.maxOrNull { it.width } ?: 0)
        .coerceIn(constraints.minWidth, constraints.maxWidth)
        val h = ps.sumOf { it.height }
        .coerceIn(constraints.minHeight, constraints.maxHeight)
        layout(w, h) {
            var y = 0
            ps.forEach { it.placeRelative(0, y); y += it.height }
        }
    }
```

One wrapped node only? Modifier.layout { m, c -> ... } (measure it, add an offset/baseline gap). **vs SwiftUI Layout**: a proposal is a *suggestion* and a subview may be asked sizeThatFits several times; Compose constraints are hard bounds and each child is measured **once**.

SubcomposeLayout & BoxWithConstraints

- SubcomposeLayout **composes during layout** — children exist only once constraints are known (LazyColumn, Scaffold, BoxWithConstraints). Composition moves into the layout pass; a new constraint re-subcomposes.
- Use it only when the *tree* depends on size (1 vs 2 panes); to *position* by size use Layout. onSizeChanged + state write = a frame late.

Modifier.Node, not composed{ }

composed{ } runs composition per use site, allocates, never skips. A Modifier.Node lives long; its ModifierNodeElement (create, update, **equals/hashCode** — a data class) is diffed and update() mutates the node. Roles: LayoutModifierNode, DrawModifierNode, PointerInputModifierNode, SemanticsModifierNode.

```
fun Modifier.circle(c: Color) = this then CircleEl(c)
private data class CircleEl(val c: Color) :
    ModifierNodeElement<CircleNode>() {
    override fun create() = CircleNode(c)
    override fun update(node: CircleNode) { node.c = c }
}
private class CircleNode(val c: Color) :
    Modifier.Node(), DrawModifierNode {
    override fun ContentDrawScope.draw() { drawCircle(c); drawContent() }
```

Moving things & lazy lists

- Animating padding/size/offset(x.dp) → **relayout**, neighbours move. graphicsLayer { translationX; scaleX; alpha } transforms the drawn layer only — **no remeasure**; a scaled item keeps its old slot.
- key = { it.id } (Bundle-able) keeps item state across inserts; contentType reuses same-type compositions; Modifier.animateItem() (fade + placement, needs keys) replaced animateItemPlacement.

Traps

- !** LazyColumn in a verticalScroll → crash: **infinite max height**. Use one lazy list with several item{ }s.
- !** “size(50) is 50” — not after fillMaxSize(): the min wins.
- !** No margin API: padding *before* background is the margin.

Remember: constraints down, sizes up, place last · measure once · order = wrapping · move with graphicsLayer.